

Sustainable Public Debt and Economic Growth under Wage Rigidity

A review of Greiner (2013)

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Abstract

Greiner (2013) analyses the effects of public debt on long-run economic growth in an AK endogenous growth model featuring persistent unemployment due to wage rigidities. This review presents the structure of the model, its main analytical results regarding the existence and uniqueness of balanced growth paths, the neutrality of public debt, and the stability of the economy.

1. Introduction

A central question in macroeconomics concerns the long-run effects of public debt on economic growth. The conventional wisdom, derived from models with perfect labour markets, is that higher public debt crowds out private capital accumulation, thereby reducing long-run growth. Greiner (2013) challenges this view by asking whether the same result holds when labour markets are imperfect — that is, when wage rigidities give rise to persistent unemployment. This question is of particular relevance for European economies, where unemployment has historically been a long-run structural phenomenon rather than a purely cyclical one.

The literature on endogenous growth models typically abstracts from unemployment, yet several contributions challenge this. Pissarides (1990) identifies a capitalisation effect linking growth to labour market tightness; Aghion and Howitt (1994) show how Schumpeterian creative destruction generates transitory unemployment; and Acemoglu (1997) demonstrates that strategic complementarities between firms can produce an unemployment trap. On the debt side, Greiner (2008) and Greiner (2011) show that under flexible wages, higher debt ratios reduce balanced growth rates and that a balanced budget suffices for stability. Greiner and Flaschel (2010) suggest numerically that this result reverses under wage rigidities and productive public spending — but analytical results could not be derived.

Greiner (2013) fills this gap with an analytically tractable AK endogenous growth model

featuring micro-founded wage rigidities, following Raurich et al. (2006), and a government subject to an intertemporal budget constraint. The paper asks: does public debt affect long-run growth and employment when labour markets exhibit persistent wage rigidities? The central finding is that public debt is neutral in the long run — it affects neither the balanced growth rate nor the unemployment rate — because the key driver of growth is the wage-to-capital ratio, governed by union behaviour rather than fiscal policy. Debt policy does, however, affect macroeconomic stability: insufficient fiscal reaction to rising debt can set the debt-to-GDP ratio on an explosive path.

2. Methodology

2.1 The model

The economy is populated by households, firms, a government, and trade unions operating in continuous time, with no population growth. Households maximise discounted logarithmic utility over an infinite horizon, supply labour inelastically, and hold physical capital K and government bonds B (wealth $W = K + B$). The no-arbitrage condition between the two assets yields the bond rate

$$r_b = \frac{r(1 - \tau) - \delta}{1 - \tau}, \quad (1)$$

and utility maximisation delivers the standard consumption Euler equation:

$$\frac{\dot{C}}{C} = r(1 - \tau) - \rho - \delta. \quad (2)$$

Firms use an AK production function $Y = AK(L^d)^\alpha$, $\alpha \in (0, 1)$, so that profit maximisation yields labour demand as a decreasing function of the wage-to-capital ratio $x \equiv w/K$:

$$L^d = A^{\frac{1}{1-\alpha}} \alpha^{\frac{1}{1-\alpha}} \left(\frac{w}{K} \right)^{\frac{\alpha}{\alpha-1}}. \quad (3)$$

Wage rigidities arise from union behaviour following Raurich et al. (2006): unions maximise a concave function of the wage gap $(1 - \tau)w - w^s$ weighted by labour demand, where the reference wage w^s tracks past average labour income with exponentially declining weights. This micro-foundation yields the dynamic wage equation

$$\frac{\dot{w}}{w} = \beta_L \left(\frac{L^d}{L} - 1 \right), \quad (4)$$

where $\beta_L > 0$ captures labour market flexibility and depends on union preferences γ , the tax rate τ , the replacement rate λ , and the adjustment speed θ .

The government runs a deficit rule: public debt evolves as $\dot{B} = r_b B - (1 - \tau)Y - S$, where the primary surplus follows the feedback rule

$$S = \phi Y + \beta B, \quad \beta > 0. \quad (5)$$

A strictly positive β is necessary and sufficient for intertemporal solvency (Greiner, 2008, 2011). The full dynamics reduce to a three-dimensional autonomous system in the normalised variables $(c, x, z) \equiv (C/K, w/K, B/K)$.

2.2 Balanced growth paths

A Balanced Growth Path (BGP) is defined as a trajectory along which C , K , and w all grow at the same strictly positive constant rate g , while B either grows at rate g or remains constant. Normalising by K maps each BGP to a rest point of the (c, x, z) system.

Proposition 1 establishes that labour market flexibility is the key determinant of existence and uniqueness: a unique BGP exists if and only if $\beta_L \bar{L} > \rho + \delta$; otherwise two BGPs or none exist. When two BGPs coexist, the low-growth equilibrium constitutes an underdevelopment trap. Both τ and λ reduce β_L by widening the wage gap, making multiplicity more likely.

Proposition 2 — the central result — states that debt policy does not affect the BGP for a given tax rate. The mechanism is transparent: the marginal product of capital depends on L^d , which is itself a function of $x = w/K$; since fiscal policy leaves x unchanged, it has no effect on investment incentives or long-run growth. This contrasts sharply with Greiner (2011), where a higher debt ratio always lowers the balanced growth rate under flexible wages. Neutrality hinges critically on τ being constant: endogenising τ would reintroduce a distortionary channel.

Proposition 3 characterises the steady-state debt-to-GDP ratio under permanent deficits:

$$\left(\frac{B}{Y}\right)^* = -\frac{\phi}{\rho - \beta}. \quad (6)$$

A positive debt ratio requires $\phi > 0$ when $\beta < \rho$, and $\phi < 0$ when $\beta > \rho$ to prevent the government from accumulating net wealth.

2.3 Stability of the economy

Local stability is assessed via the eigenvalues of the Jacobian of the normalised system at each rest point. Under a balanced budget (Proposition 4), the unique BGP is saddle-point stable (two negative, one positive eigenvalue), so a continuum of initial (C, w) pairs converges to it given initial debt. When two BGPs coexist, the high-growth path is saddle-point stable; the low-growth path is either asymptotically stable or has a one-dimensional stable manifold, making convergence to it unlikely.

Under permanent deficits, Proposition 5 shows that raising β increases stability by raising the number of eigenvalues with negative real parts — a consequence of β raising the determinant and lowering the trace of the Jacobian. Economically, a government that reacts insufficiently to rising debt risks an explosive debt-to-GDP path, destabilising the whole economy. Greiner notes that this qualitative result mirrors that of models with flexible wages (Greiner, 2008), suggesting fiscal prudence is a robust stabilising force regardless of labour market structure.

3. Discussion

In this section we examine the main assumptions of the model, discuss their plausibility, and assess the empirical support for the paper’s central findings.

The first critical assumption is that public spending is *unproductive*. Greiner explicitly assumes that government consumption C_p is neither productive nor welfare-enhancing. This is clearly a modelling choice designed to isolate the pure effect of debt from any confounding effect of public investment. As Greiner himself acknowledges, when public spending is productive, higher debt-financed expenditure can reduce unemployment and generate positive growth externalities, as shown numerically in Greiner and Flaschel (2010). The neutrality result of Proposition 2 should therefore be interpreted cautiously: it holds specifically in an economy where debt does not finance growth-enhancing public investment.

The second assumption concerns the tax rate. Greiner assumes that τ is constant and does not respond to changes in public debt. This is precisely what drives the neutrality result: since τ is fixed, higher debt does not distort saving or investment at the margin. Neutrality breaks down as soon as τ rises with B , since the income tax directly affects the return to capital. The assumption of a constant tax rate may be reasonable in the short run, but is difficult to defend over the long run in economies where fiscal consolidation has historically relied on tax increases.

The third assumption is that wage rigidities are modelled through trade union behaviour following Raurich et al. (2006). This provides genuine micro-foundations for the wage dynamic $\dot{w}/w = \beta_L(L^d/L - 1)$. Nonetheless, the model retains simplifications: there is no bargaining between unions and firms, unions set wages unilaterally, and a single representative union rules out strategic interactions across levels of the bargaining hierarchy. These features abstract from the institutional diversity of wage-setting across countries that Blanchard and Wolfers (2000) identify as a key factor explaining differences in European unemployment rates.

A fourth remark concerns the absence of any role for monetary policy or inflation. The model is entirely real. While consistent with its long-run focus, this means it cannot

address the interaction between debt sustainability and inflation that was central in policy debates in the post-pandemic surge in public deficits.

On the empirical side, the finding that $\beta > 0$ is necessary for stability receives strong support from the data. Bohn (1998) shows, using US data, that the primary surplus responds positively and significantly to the debt-to-GDP ratio. Greiner et al. (2007) replicate this for a panel of euro area countries, and Fincke and Greiner (2011) confirm it using time-varying parameter estimates, though they find β can turn negative during episodes of fiscal stress — consistent with the model’s prediction that insufficient fiscal reaction leads to instability.

4. Conclusion

Greiner (2013) makes a valuable contribution to the literature on public debt and endogenous growth by introducing labour market imperfections in an analytically tractable framework. The central finding — that public debt is neutral with respect to long-run growth and unemployment when wages are rigid — stands in stark contrast to the results obtained in models with flexible labour markets. The mechanism is clear: in the presence of wage rigidities, it is the trade union’s wage-setting behaviour, captured by the ratio $x = w/K$, that governs the incentive to invest and hence the long-run growth rate. Debt policy cannot influence this variable and is therefore growth-neutral.

At the same time, the paper shows that fiscal policy is not irrelevant. The reaction coefficient β plays a decisive role in determining whether the economy converges to a balanced growth path or diverges onto an explosive debt trajectory. A government that fails to raise its primary surplus sufficiently in response to rising debt risks destabilising the whole economy, regardless of whether the labour market is flexible or not. This result is empirically supported by a broad literature estimating fiscal reaction functions, which consistently finds positive values of β in developed economies.

Two possible extensions of this work stand out. First, introducing productive public capital would allow one to analytically disentangle the growth effect of debt from the growth effect of public investment, a question that Greiner and Flaschel (2010) could only address numerically. Second, allowing the income tax rate to respond endogenously to the level of debt would break the neutrality result and bring the model closer to fiscal consolidation episodes observed in practice. These extensions would naturally come at the cost of analytical tractability, but they would substantially enrich the model’s policy implications.

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